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[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

Dec-25-0364

CSPC-514 (Software Engineering)

B.Tech. 5th (CBCS/NEP) [CSE/CSE (AI&DS)/CSE (AI&ML)]

Time : 3 Hours Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from section A, B, C and D. Section E is compulsory.

SECTION - A

1. Explain Unified Process in detail. How it overcomes the limitations of waterfall model? (12)

2. What do you understand by Evolutionary models? How Component-based development helps in software development? (12)

SECTION - B

3. What do you understand by Requirement Engineering Process? What are its different phases? Explain in detail. (12)

4. Explain the concept of conceptual model of UML. Illustrate the use of Sequence diagrams and Collaboration diagrams in detail using example. (12)

SECTION - C

5. Explain following:

   a. Black-box Testing

   b. White-box Testing

   c. Validation Testing

   d. System Testing (4×3=12)

6. What are the various Product Metrics for measuring software quality? Explain the use of metrics for analysis model and metrics for maintenance. (12)

SECTION - D

7. What do you understand by Risk Management? Explain Risk Mitigation, Monitoring and Management Plan in detail. (12)

8. What do you understand by Quality Management? Explain ISO 9000 quality standards in detail. (12)

SECTION - E (Compulsory)

9. (a) What do you understand by Software Reviews? How do they affect software quality?

   (b) Explain the concept of Rapid Software Development.

   (c) Explain few different available architectural styles of software development.

   (d) Define Debugging. Illustrate using example.

   (e) Define the term Software Measurement. How is it related with Metric?

   (f) What are Non-Functional requirements? (6×2=12)